MIL-F-15733/38H
12 January 1984
SUPERSEDING
MIL-F-15733/38G(USAF)
6 October 1980

MILITARY SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLE FL16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification sheet and the latest issue of MIL-F-15733.

Part numbers M15733/38-0001, -0002, -0004, and -0005 are inactive for new design as of the date of this specification. For new design use M15733/38-0006 for M15733/38-0001; use M15733/49-0006 for M15733/38-0002, -0004,and -0005.

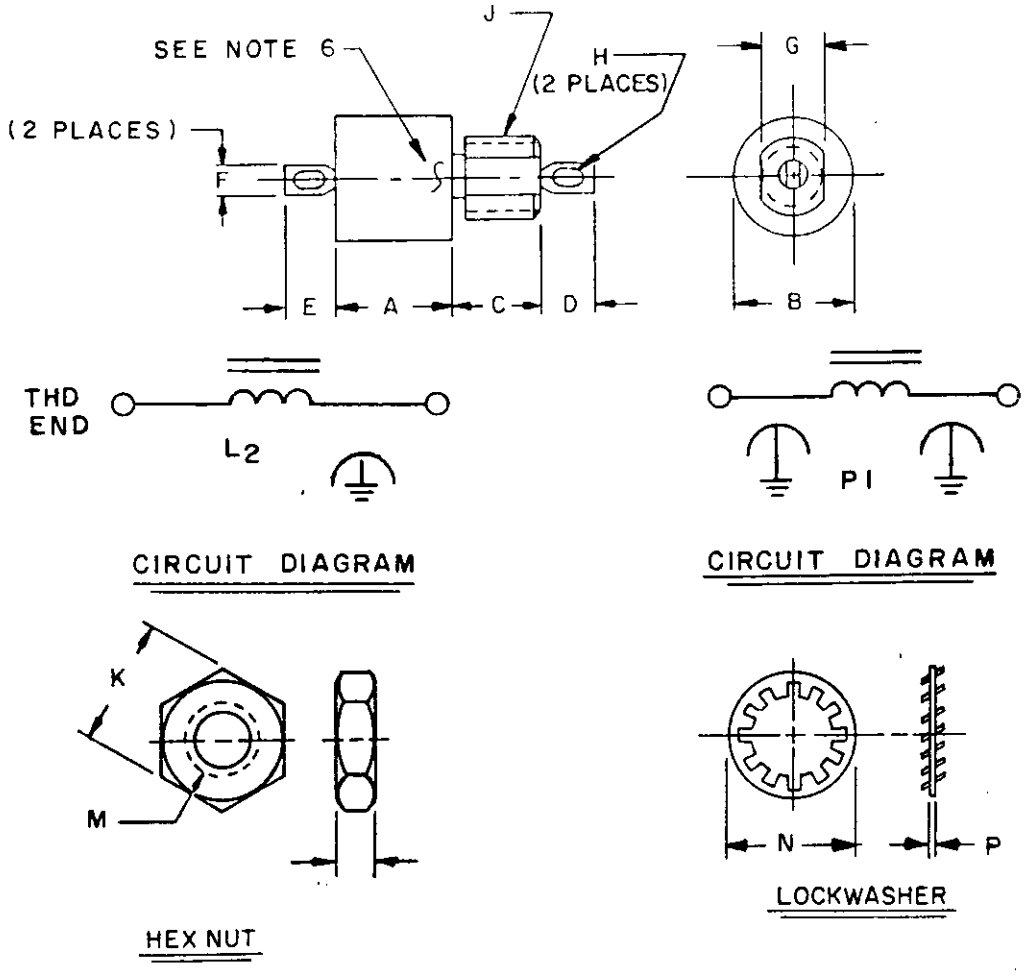


FIGURE 1. Filter and hardware dimensions, and circuit diagrams.

Ⓗ denotes changes

FSC 5915

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(H)

Filter dimensions.

Dash no.	A		B		C		D		E		F	G		H 1/		J
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	±.015	Min	Max	±.010	±.010	Mounting thread
0001	.290 (7.37)	.310 (7.87)	.38 (9.7)	.42 (10.7)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0002	.160 (4.06)	.291 (7.39)	.395 (10.03)	.415 (10.54)	.180 (4.57)	.200 (5.08)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0003	.530 (13.46)	.550 (13.97)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0004	.160 (4.06)	.180 (4.57)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0005	.160 (4.06)	.180 (4.57)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0006	.160 (4.06)	.291 (7.39)	.365 (9.27)	.415 (10.54)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0007	.210 (5.33)	.230 (5.84)	.365 (9.27)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.105 (2.67)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0008	.160 (4.06)	.180 (4.57)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.097 (2.46)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0009	.290 (7.37)	.310 (7.87)	.365 (9.27)	.385 (9.78)	.177 (4.50)	.197 (5.00)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A
0010	.290 (7.37)	.310 (7.87)	.365 (9.27)	.385 (9.78)	.302 (7.67)	.322 (8.18)	.100 (2.54)	.190 (4.83)	.100 (2.54)	.190 (4.83)	.115 (2.92)	.190 (4.83)	.210 (5.33)	.060 (1.52)	.080 (2.03)	.250-28 UNF-2A

1/ Optional slot maybe supplied .050 ±.010 inches (1.27) (.25) x .070 ±.010 inches. (1.78)

FIGURE 1. Filter and hardware dimensions, and circuit diagrams - Continued.

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(H) Hardware dimensions.

Dash no.	Hex nut					Lockwasher			
	K		L		M	N		P	
	Min	Max	Min	Max	Thread	Min	Max	Min	Max
-0001	.302 (7.67)	.322 (8.18)	.085 (2.16)	.095 (2.41)	.250-28 UNF-2B	.395 (10.03)	.408 (10.36)	.010 (.25)	.030 (.76)
-0002	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0003	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0004	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0005	.307 (7.80)	.317 (8.05)	.068 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0006	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0007	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0008	.307 (7.80)	.317 (8.05)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0009	.302 (7.67)	.322 (8.18)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)
-0010	.302 (7.67)	.322 (8.18)	.088 (2.24)	.098 (2.49)	.250-28 UNF-2B	.395 (10.03)	.405 (10.29)	.019 (.48)	.025 (.64)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardware.
5. Terminal thickness shall be .021 (0.53 mm)/.011 (0.28 mm).
6. Terminal identification. The case shall be marked at the threaded end of the filter, with the symbol "L", for dash numbers 0001, 0002, 0004, 0005, 0006, and 0008, 0009, and 0010.

FIGURE 1. Filter and hardware dimensions, and circuit diagrams - Continued.

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Ⓗ TABLE I. Electrical characteristics.

Dash no.	Circuit configuration	Max rated current (amps)	Rated DC voltage (volts)	Rated AC 2/ voltage (volts rms)	Voltage drop (volts dc max)	Minimum full load insertion loss (dB) in accordance with MIL-STD-220 at +25°C 1/														
						30 kHz	100 kHz	150 kHz	200 kHz	300 kHz	500 kHz	1 MHz	2 MHz	5 MHz	10 MHz	20 MHz	50 MHz	100 MHz	1 GHz	10 GHz
0001 3/	L_2	10	50	-	.040	-	24	-	32	-	39	44	50	57	64	55	55	55	-	-
0002 3/	L_2	15	50	-	.12	15	-	28	-	33	-	44	-	-	64	-	-	-	70	70
0003	Pi	3	150	-	.15	-	-	7	-	24	-	54	-	-	80	-	-	80	80	80
0004 3/	L_2	10	50	-	.08	15	-	28	-	33	-	44	-	-	60	-	-	-	70	-
0005 3/	L_2	10	50 4/	-	.08	15	-	28	-	33	-	44	-	-	60	-	-	-	70	-
0006	L_2	15	50	-	.12	15	-	28	-	33	-	44	-	-	64	-	-	-	70	70
0007	Pi	15	50	-	.09	70 dB from 420 MHz to 1.0 GHz, inclusive														
0008	L_2	15	200	125	.12	-	-	-	-	-	-	6	-	-	25	-	-	-	60	-
0009	L_2	15	250	125	.09	4	-	15	-	21	-	31	-	-	50	-	-	-	70	80
0010	L_2	15	250	125	.09	4	-	15	-	21	-	31	-	-	50	-	-	-	70	80

1/ Full-load insertion loss measurements shall be performed over the frequency range of 100 kHz to 20 MHz inclusive. Measurements below or above this frequency range shall be performed at no-load.

2/ From dc to 400 Hz.

3/ Inactive for new design.

4/ Voltage rating at +125°C: (Note - derated from 80 V dc at +85°C) for dash number 0005.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Ⓗ Weight: 5 grams maximum.

Case: Metal.

Seal: Glass-to-metal.

Terminals: Solderable (see figure 1).

Operating temperature range:

Ⓗ Dash 0001 thru 0006, 0008, 0009 and 0010: -55°C to +125°C.

Dash 0007: -55°C to +85°C.

Rated voltage: See table I.

Rated current: See table I.

Creepage and clearance: Not applicable.

Ⓗ Seal: Method 112, MIL-STD-202, test condition A or D.

Capacitance to ground: Not specified.

Temperature rise: 25°C maximum.

Ⓗ Dielectric withstanding voltage:

Test voltage: 2.5 times rated dc voltage.

Maximum charging current: 50 milliamperes.

Insulation resistance: Method 302, MIL-STD-202.

Test temperature: +25°C.

Ⓗ Test potential: Rated dc voltage or 100 volts dc whichever is less.

Ⓗ Maximum charging current: 50 milliamperes.

Insulation resistance shall be not less than:

Dash numbers	Megohms
0001	500
0002 thru 0007, 0009, and 0010	100
0008	1,000

Voltage drop: See table I.

Ⓗ Insertion loss:

At 25°C: Shall be as specified in table I.

AT -55°C and at the maximum operating temperature: Shall be as specified in table I except as follows:

Up to 10 MHz: A degradation of 4 dB from the value specified in table I shall be allowed.

Ⓗ Terminal strength: Method 211, test condition A (pull), MIL-STD-202. Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

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Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition A, MIL-STD-202.

Thermal shock and immersion:

Thermal shock: Method 107, test condition A, MIL-STD-202, except the temperature at step 3 shall be at +125°C, (dash 0001 thru 0006, 0008, 0009, and 0010).

Immersion: Method 104, test condition A, MIL-STD-202.

Measurements after final cycling:

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds. Maximum charging current: 50 mA.

Insulation resistance shall be not less than 30 percent of the initial requirements.

Insertion loss (check test) shall be measured and shall meet the initial requirements.

Ⓗ Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Electrical load: 100 percent of the rated voltage between terminals and case. After the test, insertion loss (check test) shall be measured and shall meet initial requirements.

Ⓗ Vibration, high frequency: Method 204, test condition D (20 g), MIL-STD-202.

Electrical load: Rated dc voltage and current shall be applied continuously during test.

Moisture resistance: Method 107, test condition A, MIL-STD-202, except that the emperature at step 3 shall be the maximum operating temperature. Filters shall then be tested in accordance with method 106 of MIL-STD-202.

Polarization voltage: Rated dc voltage or 100 volts dc whichever is less.

Life: In accordance with MIL-F-15733 and method 108 of MIL-STD-202:

Qualification inspection: Test condition D (1,000 hours).

Group C inspection:

Dash numbers	Test condition
0001, 0002, 0004 thru 0010	B (250 hours)
0003	D (1,000 hours)

Part number: M15733/38- (dash number from table I).

Marking: Marking shall be as follows:

For dash numbers 0001 thru 0010, the filter shall be marked with the following minimum information:

a. Military part number.

b. Source code.

c. Date code.

d. Terminal identification.

In addition, full marking in accordance with MIL-F-15733 shall be marked on the unit package. For dash nmber 0003 the filter shall be marked with full marking in accordance with MIL-F-15733.

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Air Force - 11, 17, 99
DLA - ES

User activities:
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Navy - MC, AS, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-15733/38H	Filters, Radio Frequency Interference, Hermetically Sealed, Style FL16

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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